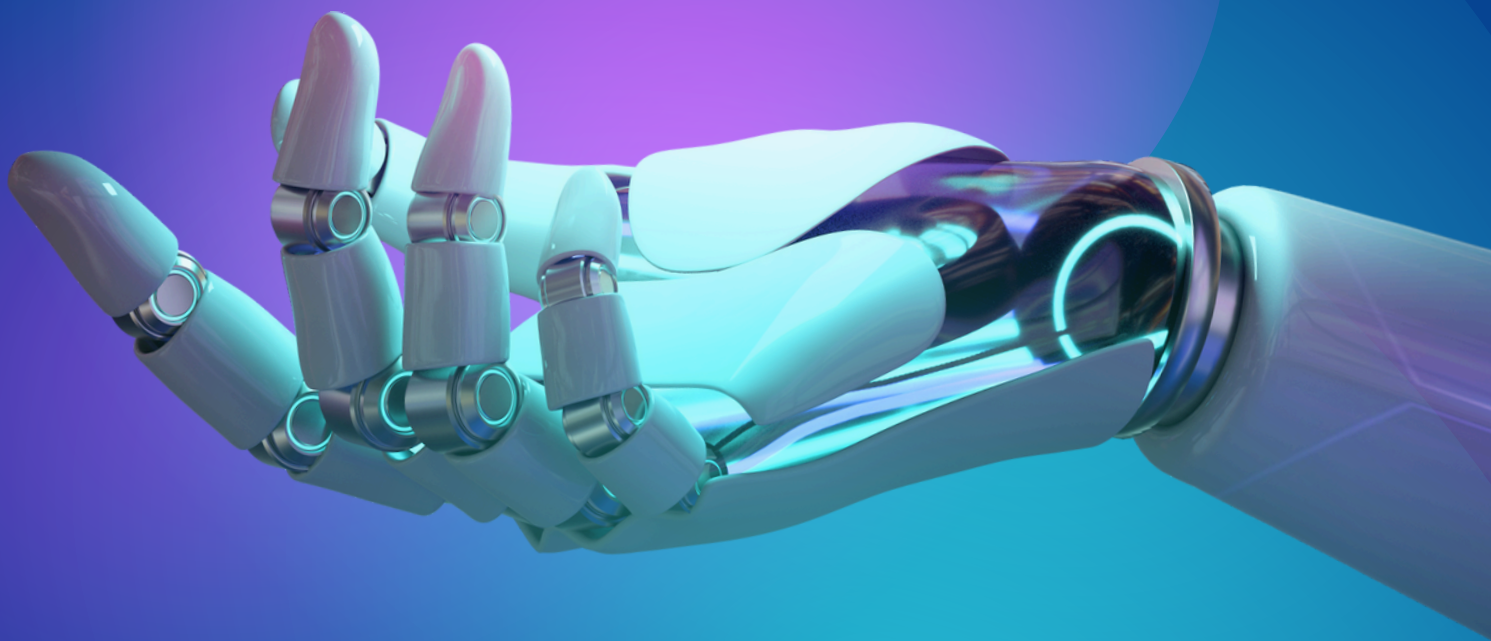


PROGRAMMING 3C

P.O.E PART 1

PLANNING & DESIGN



ST10252746

THESHARA NARAIN

PROG7313

P.O.E PART 1**PLANNING AND DESIGN****AC. Tech:**

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Introduction

According to research depicted in Botsi (2024), the lack of effective financial management often leads to poor financial well-being, a lot of people struggle to make and stick to budgets or keep track of their spending habits. This challenge can lead to financial instability, missed savings goals, and an inability for people to achieve long-term financial security. To address these issues, AC. Tech, our team of developers for the personal finance management application, named BudgetBuddy, created a user-friendly, automated, and interactive budgeting experience. By drawing inspiration from leading budgeting applications such as YNAB by Jesse Mecham, Money Tracker by HoroScope365, and Wallet by BudgetBakers, this document and its research describes the design, functionality, stand-out features, and development approach of creating this application. BudgetBuddy stands out in the mobile-application market because of its real-time modification of budgets, financial analytics, gamification features meant to promote sensible spending patterns, visual analytics that lets users monitor spending, create savings targets, handle regular payments, and get insightful financial knowledge. This document outlines a well-organized and user-centred approach for developing BudgetBuddy. It includes essential details such as the app's fundamental requirements, user interface design, innovative features aimed at enhancing user experience, and a thorough project plan that specifies the development timeline.

App Overview

BudgetBuddy is a mobile application that simplifies personal finance management through expense tracking, budgeting, and financial goal management. It features a clean, approachable interface with a mint green colour scheme, allowing users to create customized expense categories, set monthly budget limits, and log transactions. The homepage provides a quick overview of budget status, spending trends, and category-wise expenditure breakdowns. Users can track spending patterns through visualizations and earn badges for consistent expense logging. The app also incorporates gamification elements, making financial discipline more engaging. Users can customize categories with images or icons for visual impact. The Budget Assistant chatbot offers financial guidance and assistance. The app also offers a smart transaction search, detailed expense listings, and savings tracking feature.

App Name: BudgetBuddy

Icon Design:



Innovative Features:

- **Scheduled Payments:** Users can automate recurring expenses like rent, bills, and subscriptions, ensuring payments are made on time without manual entry. This helps with better financial planning and avoids missed payments (Horoscope365, 2020).
- **Reminders:** The app provides automatic reminders for upcoming bills. Users receive notifications before due dates, helping them avoid late fees and penalties. These reminders ensure that even non-automated expenses, such as one-time payments, are not forgotten, making financial management more reliable (Horoscope365, 2020).
- **Interactive Visual Budgeting & Financial Insights:** The app combines color-coded progress bars (yellow for incomplete, green for completed, red for overspending) with detailed spending reports and net worth graphs. This real-time visual tracking and financial analysis help users monitor budgets, identify trends, and make informed financial decisions (Business Insider, 2024).
- **Multi-Profile User Support:** Whether personal, family, or company budgeting, the application lets users establish and move between many different accounts, therefore meeting various financial demands. Every profile guarantees a customized experience for people and businesses equally by including their own sources of income, cost categories, and financial objectives (BudgetBakers, 2024).
- **Smart Transaction Search & Filters:** The app allows users to search for expenses using keywords, dates, and amounts. Makes it easy to find specific transactions quickly. Add advanced filtering options (for example: Find all transactions over R50 in the last 3 months) (Horoscope365, 2020).
- **Multi-Currency & International Support:** The app allows users to manage multiple currency accounts, making it useful for travellers and international users (BudgetBakers, 2024).

App Requirements

Minimum Requirements

- **User Registration & Authentication:** Users must be able to create an account (Register) and securely log in using their email and password. Firebase Authentication will be used to manage secure login, user registration, and password recovery. Users will also log out at any time to maintain privacy and security.
- **Expense Categories Management:** Users should be able to create, edit, and delete categories for their expenses, such as Groceries, Transport, or Entertainment. These categories will be stored in RoomDB for persistence, ensuring they remain available even when the user is offline. Additionally, users will have the option to set a budget limit for each category to manage their spending effectively.
- **Expense Entry System:** Users must be able to add expense entries with key details, including the amount, date (defaulting to the current date but allowing changes), a description, and a selected category. The system should also allow users to edit or delete expenses. All expense data will be stored in RoomDB for offline access, ensuring users can track their spending even without an internet connection.
- **Expense Receipt Upload:** Users will have the ability to attach a receipt photo to each expense entry for better financial tracking. The receipt images will be stored locally, making them easily accessible from the expense list whenever needed.
- **Budget Management:** Users can set a total monthly budget goal to track their overall spending. A visual progress meter will indicate how much of the budget has been used. If users exceed their budget, a warning notification will alert them, helping them stay within their financial limits.
- **Expense List & Filtering:** The app will provide a complete list of all expenses, allowing users to review their spending history. Users will also have the option to filter expenses based on specific criteria, such as date range, category, and amount range, making it easier to find relevant transactions.
- **Category-Based Spending Overview:** Users will be able to view their total spending per category within a selected period. To enhance usability, spending trends will be displayed visually using bar graphs or pie charts, helping users understand their spending habits more clearly.
- **Progress Dashboard (Final PoE Requirement):** A dedicated progress dashboard will provide an overview of the user's spending progress for the current month. It will visually highlight overspending categories, making it easier for users to identify areas where they need to cut down on expenses.
- **Graphical Spending Insights (Final PoE Requirement):** Users will have access to a graphical representation of their daily spending trends. The graph will show category-wise spending over a selected period, helping users analyse their financial patterns and make informed decisions.

- **Cloud Data Storage (Final PoE Requirement):** User data will be synced to Firebase Firestore, ensuring multi-device access and data backup. Users will also be able to restore their data from the cloud, providing security and continuity even if they switch devices or reinstall the app.

Innovative Requirements

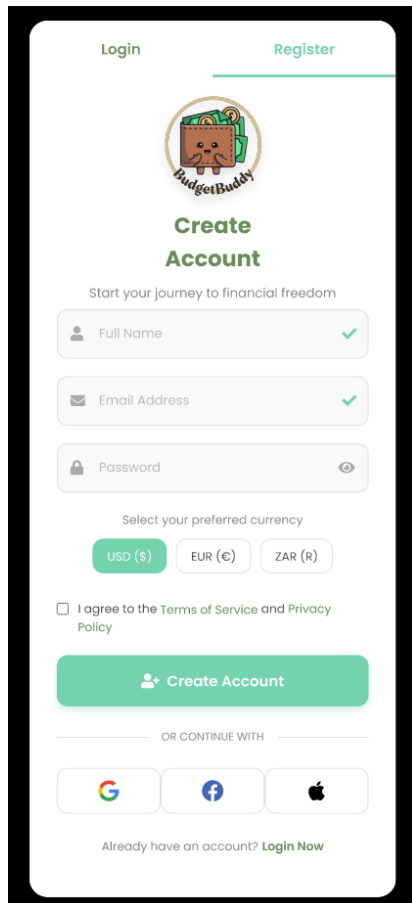
- **Feedback system for users:** The feedback system is designed to give users a way to engage with the app and receive prompt responses to their inquiries. This could be set up as a chatbot that gives automated answers to common user questions, or as a Q&A section where users can look through frequently asked questions (FAQs) and discover useful articles, tips, and guides. This feature aims to make sure users feel backed up and in the loop, improving the overall experience by giving them quick access to information and help with troubleshooting (Piluta, 2021).
- **Personalize User Profile Picture:** The app will allow for a more personalized user experience; users will have the option to upload a custom display picture (profile picture) or select from a pre-provided collection of images. This feature allows users to express their individuality within the app, making the experience feel more personal. The ability to choose a display picture can make users feel more connected to the app, encouraging frequent use and engagement (Piluta, 2021).
- **Customized Category pictures:** Like the profile picture feature, users will be able to upload custom images for their expense categories (for example: Groceries, Transport) or select from a library of images provided within the app. This feature enables users to visually represent each category in a way that resonates with them, helping to organize their expenses in a more intuitive and personalized manner. This will enhance the user interface, making it visually appealing and user-friendly (Piluta, 2021).
- **Introduce a Dark Mode Scheduler:** The Dark Mode Scheduler will switch between light and dark modes based on time of day or user preferences. For instance, dark mode could be enabled at night for easier viewing in low-light environments, while light mode is used during the day for better visibility in bright settings. Users can also manually toggle this feature to suit their preferences. This feature enhances the user experience by offering a comfortable viewing environment that reduces eye strain in different lighting conditions, ensuring the app remains user-friendly at all times of day (Piluta, 2021).
- **Saving Tracking:** To encourage users to save money, a saving tracking feature will allow users to set specific savings goals (for example: saving R500 by the end of the month). A progress meter or challenge ring will visually display the progress toward the goal. As users contribute to their savings, the ring will fill up, giving them a tangible sense of achievement. Additionally, once users complete certain savings milestones, they will unlock secret achievements or rewards, adding an element of gamification to the process. This feature motivates users to keep working towards their savings goals while making the experience more engaging and fun (Piluta, 2021).

User interface design

Using Figma:

<https://www.figma.com/design/4EZxWxwvW6NwEdt7YOERMk/BudgetBuddy?node-id=0-1&t=gfZVly5cPQxH6Alu-1>

Registration screen:

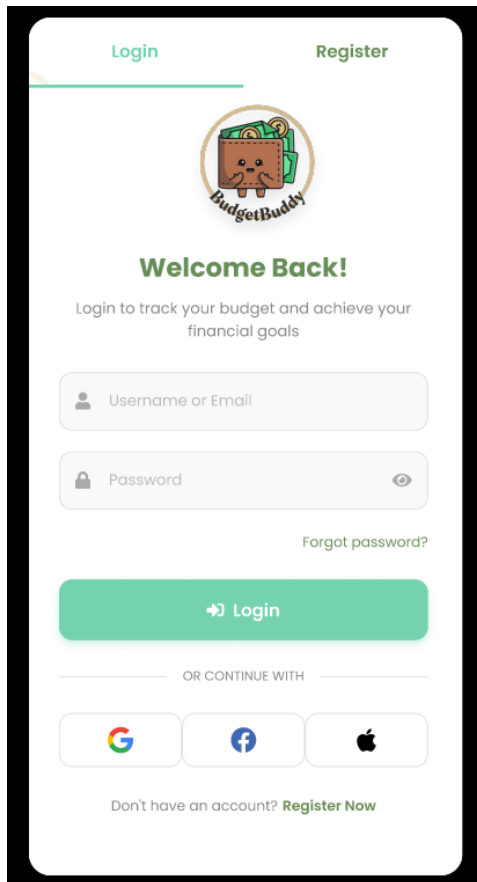
The image shows a mobile app registration screen for 'BudgetBuddy'. At the top, there are two tabs: 'Login' and 'Register', with 'Register' being the active tab. Below the tabs is the 'BudgetBuddy' logo, which features a cartoon piggy bank character. The main heading is 'Create Account' in green, followed by the tagline 'Start your journey to financial freedom'. The registration form consists of three input fields: 'Full Name' (with a green checkmark), 'Email Address' (with a green checkmark), and 'Password' (with an eye icon for toggling visibility). Below these fields is a section for 'Select your preferred currency' with three buttons: 'USD (\$)' (highlighted in green), 'EUR (€)', and 'ZAR (R)'. A checkbox labeled 'I agree to the Terms of Service and Privacy Policy' is present. A large green button with a plus icon and the text 'Create Account' is positioned below the checkbox. Underneath this button is the text 'OR CONTINUE WITH' followed by three social media login options: Google, Facebook, and Apple. At the bottom, there is a link that says 'Already have an account? Login Now'.

The registration screen is where new users can set up an account on the BudgetBuddy app. To make sure that the sign-up process is secure, it prompts for the user's full name, email address, and password. Users can also select which currency they would like to use, which makes the banking experience fit their varying needs. Terms of service and privacy rules, which are needed to make an account, are also shown on the screen. For quick sign-up, Google, Facebook, and Apple all offer different sign-up methods. People who already have an account can click the "login now" option to login with their existing credentials.

Features satisfied:

- **POE requirement:** Login and register
- **Budget Buddy Innovative requirement/feature:** Multiple currency selection

Login screen:



Existing users are able to access their BudgetBuddy accounts by entering their username or email and password on the login interface. The app offers users convenience by supporting alternative sign-in systems such as Google, Facebook, and Apple, and it includes a “Forgot Password?” option for account recovery. The interface also provides a direct link to the signup page for new users who have not yet registered.

Features satisfied:

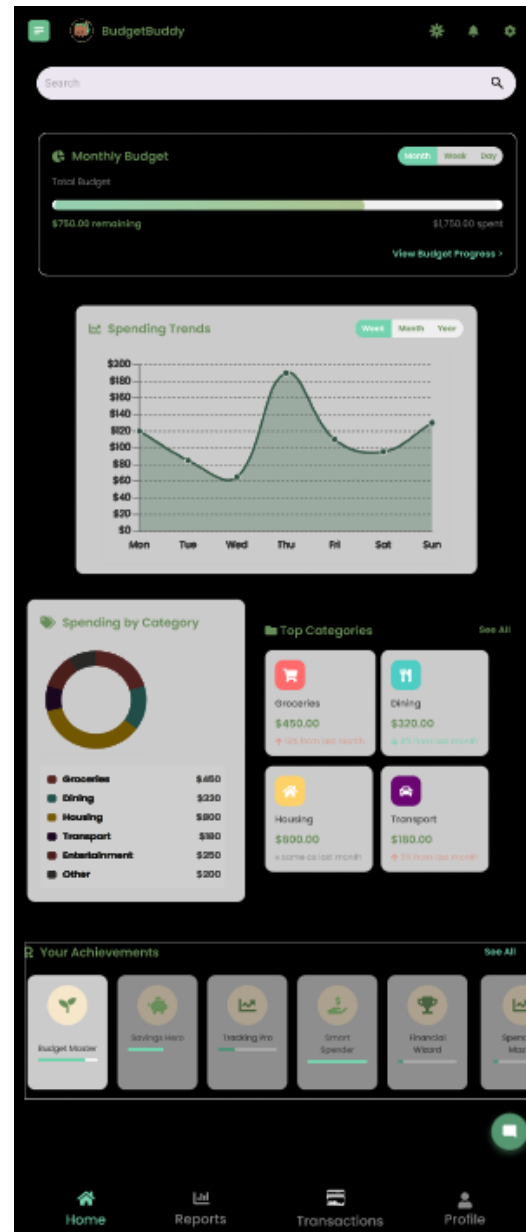
- **POE requirement:** Login and register

Home Dashboard screen:

(Light and Dark Mode)



Screen 1: Light Mode



Screen 2: Dark Mode

Screen 1: The scrollable main dashboard page displays users' monthly budgets, spending by category, and spending patterns. It provides them with a comprehensive understanding of their financial situation. It provides current charts that can assist in monitoring spending patterns, as well as key areas of spending and percentage changes from the previous month. Users can also view their financial achievements, which promotes engagement and consistency. The page also features a navigation bar that allows users to switch between the applications main pages. Screen 2: This screen displays the dark mode function, which offers an alternative theme for those who prefer a darkened interface. Dark mode is particularly beneficial in reducing eye strain in low-light environments. Users have the option to switch between light and dark themes, based on their preferences.

Features satisfied:

- **POE requirement:** The user must be able to view a graph showing daily spending over a user-selectable period.
- **POE requirement:** The app must include gamification elements such as rewards or badges for meeting budget goals or consistent expense logging (final PoE only).
- **Budget Buddy Innovative requirement/feature/feature:** Feedback system for users - The feedback system is designed to give users a way to engage with the app and receive prompt responses to their inquiries. This is set up as a chatbot that gives automated answers to common user questions, or as a Q&A section where users can look through frequently asked questions (FAQs) and discover useful articles, tips, and guides.
- **Budget Buddy Innovative requirement/feature:** Introduce a Dark Mode Scheduler - The Dark Mode Scheduler will switch between light and dark modes based on time of day or user preferences.

Add Expense screen:

The screenshot displays the 'Add Expense' interface. At the top, there's a title bar with a back arrow, the text 'Add Expense', and a moon icon. Below this, a large green '\$' symbol is next to the '0.00' total amount. A numeric keypad with digits 1-9, a decimal point, and a zero key is provided. A 'Hide Keypad' link is below the keypad. The 'Date' section includes a text input 'mm/dd/yyyy', a calendar icon, and a 'Today' button. The 'Category' section has a text input 'Select or type to search'. The 'Description (Optional)' section has a text input 'Ex: Dinner with friends'. The 'Receipt (Optional)' section has a camera icon and 'Attach receipt photo'. A green 'Save Expense' button with a checkmark is at the bottom.

This screen ensures that users' financial behaviour is accurately monitored by allowing them to directly record their expenses. Users have the ability to input the quantity they paid, select the category such as food, rent, or entertainment, and record the date of the transaction. Users can also optionally attach their receipt for additional information or reference.

Features satisfied:

- **POE requirement:** The user must be able to add an expense entry, specifying at least the amount, date, description, and category.
- **POE requirement:** The user must be able to optionally attach a photo (e.g., a receipt) to each expense entry.

Budget Goals screen:

BudgetBuddy

Set Your Budget Goals

Plan your monthly spending by allocating budgets to different categories.

Set Monthly Budget

\$ 2,150.00

Suggested budget: \$2,280

1 2 3
4 5 6
7 8 9
0 . **\$**

Category Allocation [Add Category](#)

Groceries	\$500	▼
Dining Out	\$300	▼
Adjust Budget: \$300		
300		
Suggested: \$270		
Transportation	\$200	▼
Entertainment	\$150	▼
Housing	\$1000	▼

Budget Overview \$2,150 / \$2,150

\$2,150 allocated \$0 remaining

Groceries	\$500
Dining Out	\$300
Transportation	\$200
Entertainment	\$150
Housing	\$1000

[Cancel](#) [Save Budget](#)

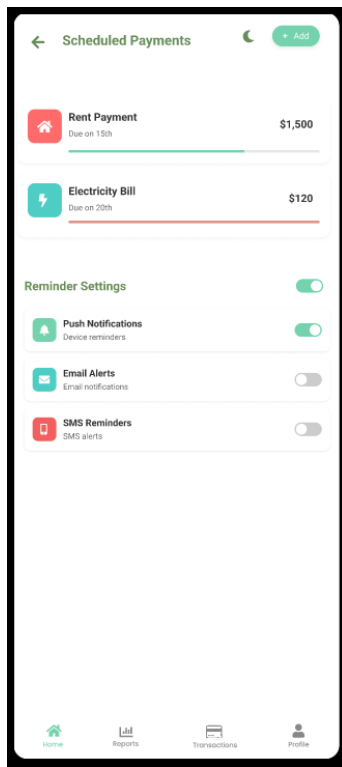
Set budget goals with category allocation and budget overview:

This screen contains the function to assist users in establishing financial objectives by enabling them to establish budgetary limitations for specific expenditure categories. Users can set up budgets for categories like groceries, dining, transportation, and other essential expenses. Furthermore, this screen contains a budget overview that tracks progress visually, the system shows how much has been spent and how much still remains in each category.

Features satisfied:

- **POE requirement:** The user must be able to set a monthly total budget goal and specific limits for individual categories.

Schedule payments notifications and reminder settings screen:

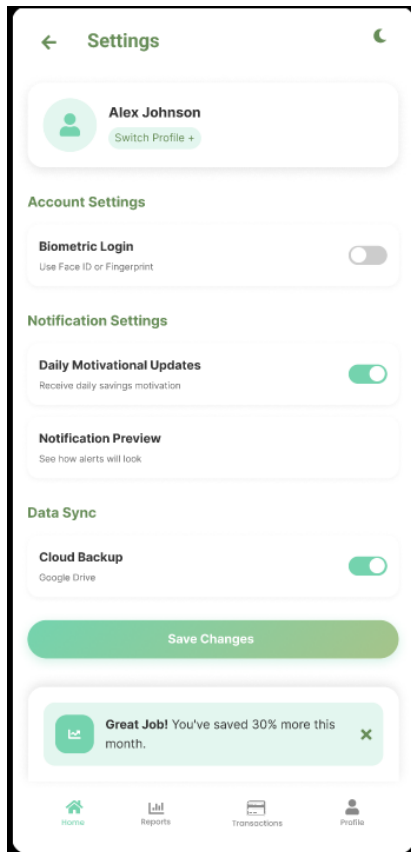


This screen lets users set up automatic and recurring payments for expenditures such as rent, bills, loans, and subscription fees. Users can ensure they never miss a payment by setting up reminders that send them messages before a due date. Users can choose from email, push messages, or in-app alerts, and they can also change how often they are reminded.

Features satisfied:

- **Budget Buddy Innovative requirement/feature:** Scheduled Payments - Users can automate recurring expenses like rent, bills, and subscriptions, ensuring payments are made on time without manual entry.
- **Budget Buddy Innovative requirement/feature:** Reminders - The app provides automatic reminders for upcoming bills.

Settings screen

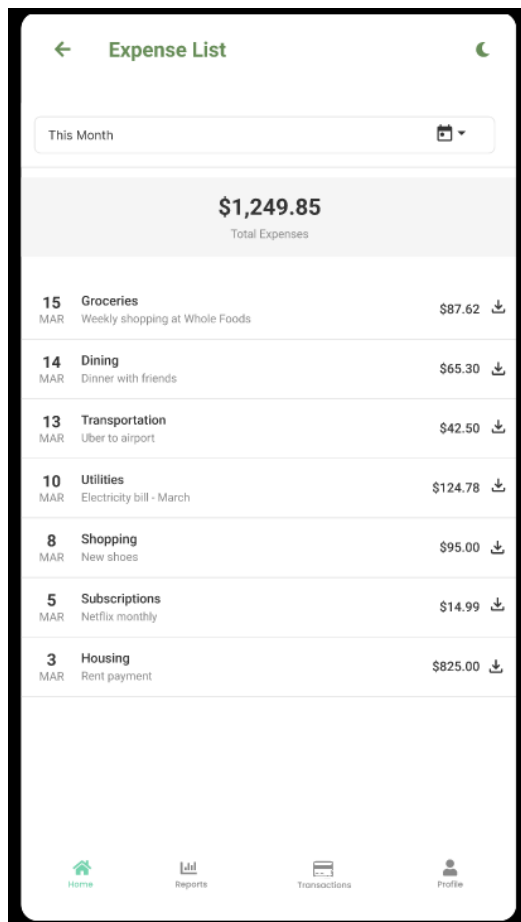


The settings screen provides users with customization options to personalise their financial management experience. Users are permitted to modify their accounts' language, currency, and protection preferences, including two-factor authentication. In addition, they have the ability to select which notifications they wish to receive, modify privacy preferences, and manage accounts that are associated with them.

Features satisfied:

- **Budget Buddy Innovative requirement/feature:** Multi-Profile User Support - Whether personal, family, or company budgeting, the application lets users establish and move between many different accounts, therefore meeting various financial demands.
- **Budget Buddy Innovative requirement/feature:** Personalize User Profile Picture - to allow for a more personalized user experience, users will have the option to upload a custom display picture (profile picture) or select from a pre-provided collection of images.

Expense list filter screen

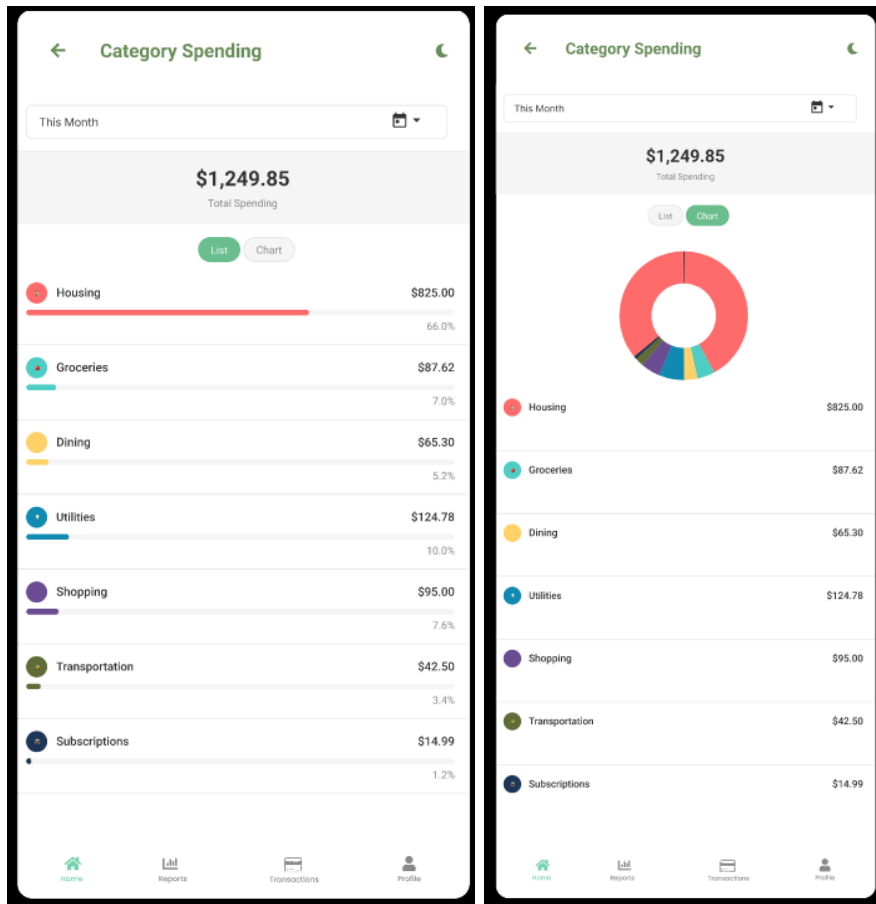


This screen lets users filter their expenditure records by certain factors, which makes it easier to keep track of their money; users can filter transactions by category, date period, or amount, which makes it easier to identify specific spending. This feature is especially useful for finding transactions that require to be examining or tracking spending trends over time.

Features satisfied:

- **POE requirement:** The user must be able to view a list of all expenses during a user-selectable period. If a receipt photo was stored for an entry, the user must be able to access it from this list.
- **Budget Buddy Innovative requirement/feature:** Smart Transaction Search & Filters- Wallet allows users to search for expenses using keywords, dates, and amounts.

Category spending screen:



Screen 1: List

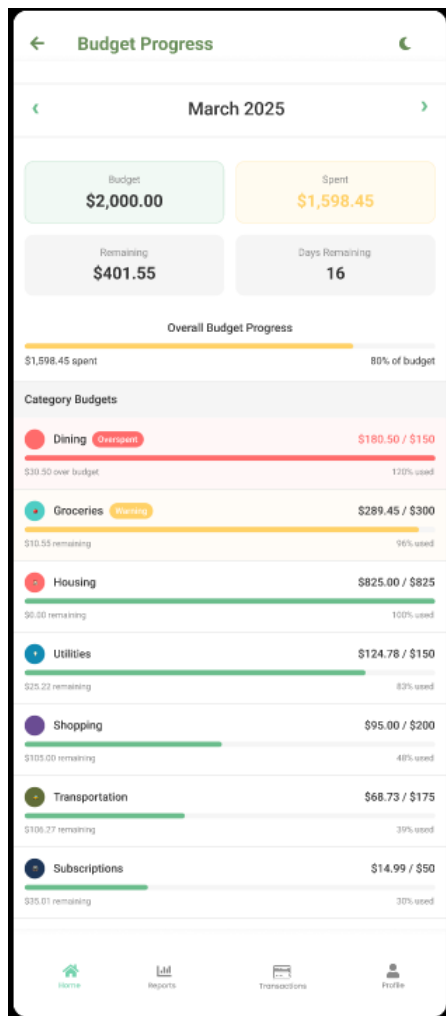
Screen 2: Chart

Screen 1: This screen organizes spending into several categories, food, transportation, entertainment, and healthcare among others. Viewing a comprehensive summary of their expenses in every category allows users to pinpoint areas in which their money is going. This function lets users compare expenditure in several areas and make changes to remain within their budget. Screen 2: This screen shows spending trends graphically, allowing the differences in costs between different categories stand out. The graph shows users where they spend the most money and where they might be able to save money. It allows for users to change their spending by looking at how much they spend over a period of weeks or months. The visual style makes it easier to read and find financial information.

Features satisfied:

- **POE requirement:** The user must be able to view the total amount spent on each category during a user selectable period.
- **POE requirement:** Interactive Visual Budgeting & Financial Insights - BudgetBuddy combines color-coded progress bars (yellow for incomplete, green for completed, red for overspending) with detailed spending reports and net worth graphs.
- **POE requirement:** The user must be able to view the total amount spent on each category during a user selectable period.

Budget progress screen:

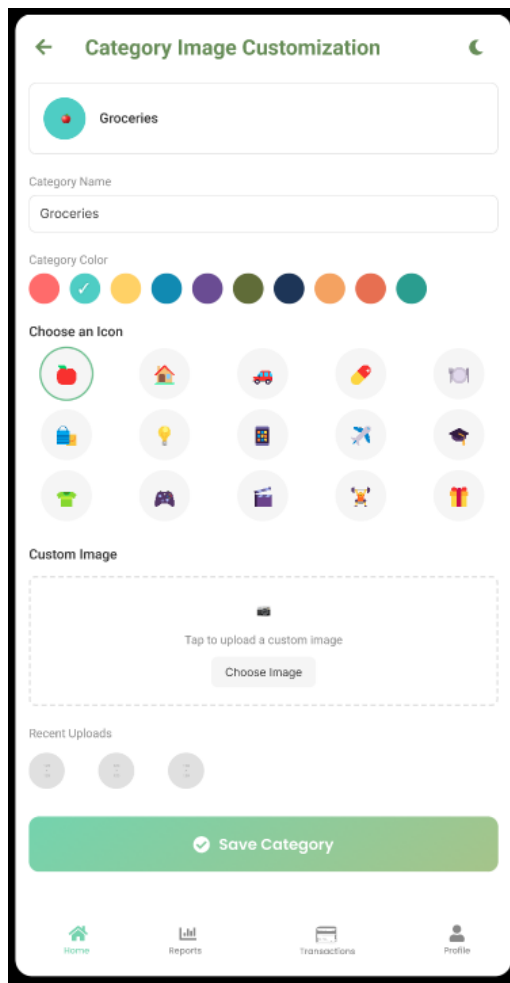


This screen guides users toward their budgets by showing them how much they've spent and how much money they have remaining for the month. A progress bar shows users how close they are to meeting their budgeting goals. Users receive notifications when they are about to spend too much, which helps them avoid overspending.

Features satisfied:

- **POE requirement:** The user must be able to view the total amount spent on each category during a user selectable period. The app must display a progress dashboard that shows how well the user is staying within their budget goals for the current month. Overspending categories must be highlighted visually (final PoE only).

Category image customization:

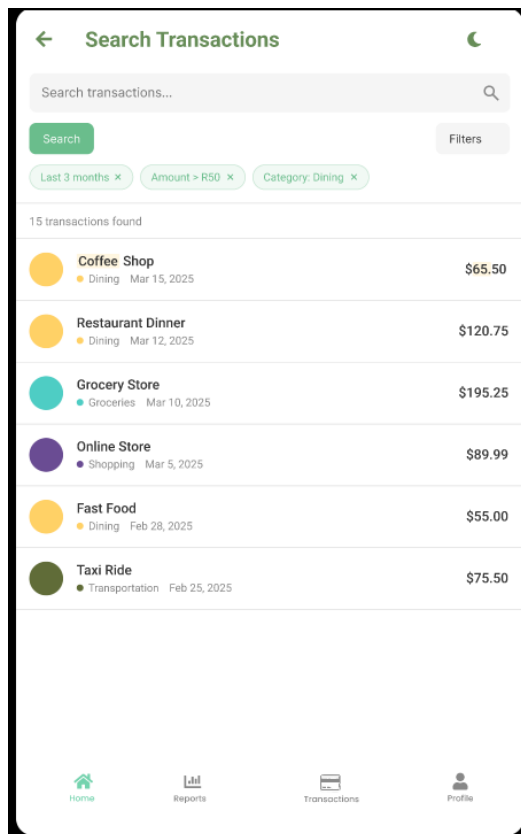


The category picture screen makes it easier for users to find what they want by visually connecting pictures with spending categories. This feature helps users quickly find expense categories, which makes the system easier to use.

Features satisfied:

- **Budget Buddy Innovative requirement/feature:** Customized Category pictures - Similar to the profile picture feature, users will be able to upload custom images for their expense categories (for example: Groceries, Transport) or select from a library of images provided within the app.

Search Transactions screen:



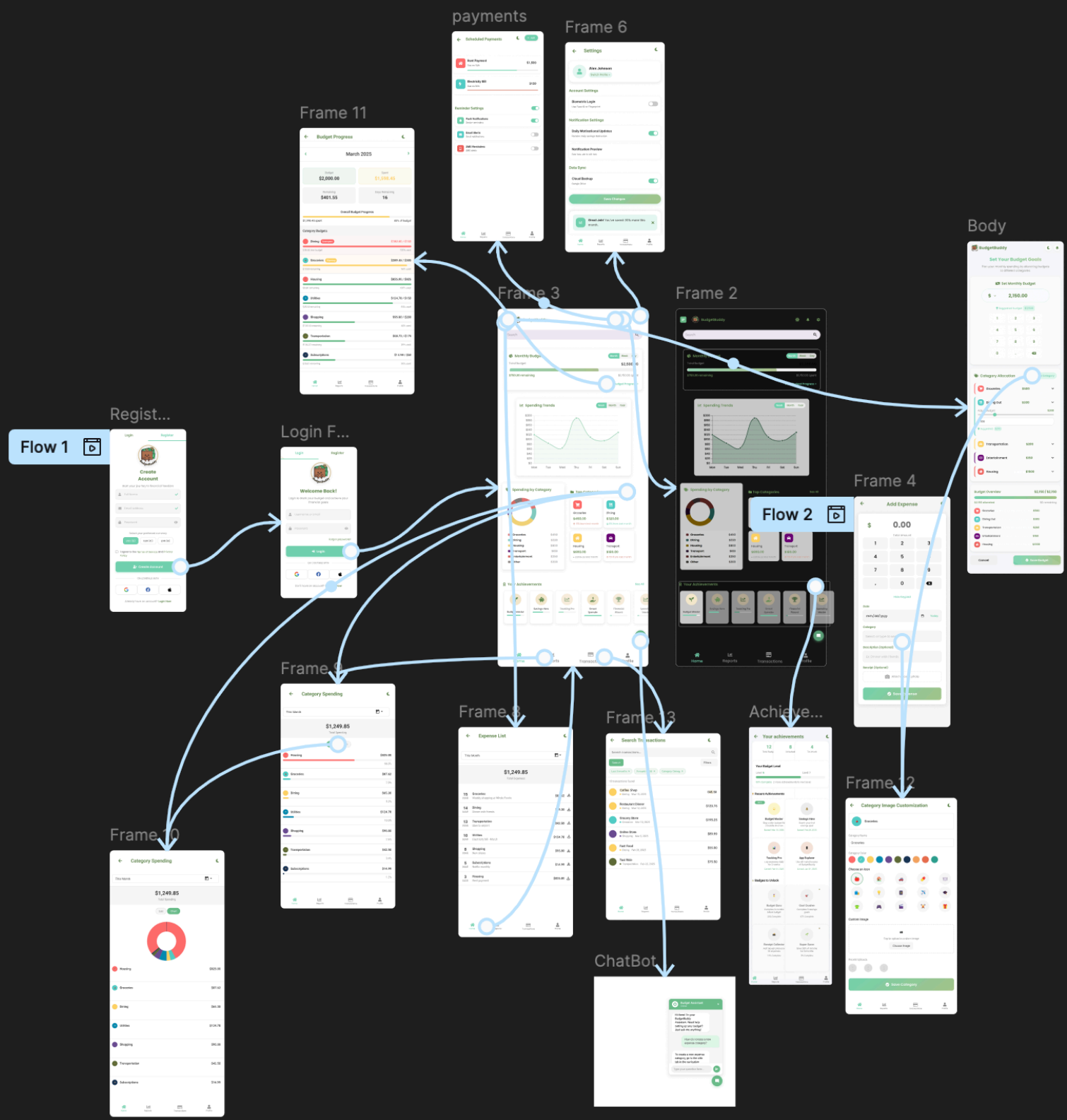
On the transactions screen, users can see a full list of all the money youth have spent, organized by date and category. Users can look back at how much they've spent in the past, keep track of financial trends, and make sure that all of their transactions are recorded correctly. Users can also look for specific transactions on this screen, which makes it easy to find purchases they've already made.

Features satisfied:

- **Budget Buddy Innovative requirement/feature:** Smart Transaction Search & Filters - Wallet allows users to search for expenses using keywords, dates, and amounts.

Navigation Diagram

<https://www.figma.com/design/4EZxWxwvW6NwEdt7YOERMk/BudgetBuddy?node-id=0-1&t=YelaQiL6zulMeiRG-1>



Project Plan

Phase 1: Research & Planning (Feb 26- Mar 16)

- **Task 1.1:** Conduct market research to analyse competitor budgeting apps, identifying strengths, weaknesses, and user needs. (Feb 26-28)
- **Task 1.2:** List and document all required features based on research findings, ensuring alignment with user expectations. (Mar 1-2)
- **Task 1.3:** Define technical and functional requirements, detailing the database structure for efficient data management. (Mar 3)
- **Task 1.4:** Design wireframes for all key screens, focusing on user experience and app flow consistency. (Mar 6)
- **Task 1.5:** Develop a project timeline with milestones, ensuring structured execution and clear progress tracking. (Mar 16)

Milestone 1: Completion of project plan, wireframes, and feature documentation by the end of Week 2.

(Lutkevich, 2021)

Phase 2: UI/UX Design (Mar 7-16)

- **Task 2.1:** Create visually appealing UI layouts for key screens using Figma, ensuring an intuitive user experience. (Mar 7-16)
- **Task 2.2:** Define colour schemes, typography, and icon styles to maintain a cohesive design language across the app. (Mar 7-16)
- **Task 2.3:** Gather user feedback on initial designs and refine the interface based on usability improvements. (Mar 13-16)

Milestone 2: Finalized UI/UX designs ready for development by the end of Week 4.

(Lutkevich, 2021)

Phase 3: Backend Development (Apr 10-27)

- **Task 3.1:** Set up Firebase Authentication to handle secure user login and signup processes. (Apr 10-Apr 12)
- **Task 3.2:** Implement RoomDB for local data storage, ensuring offline functionality and smooth user experience. (Apr 13)
- **Task 3.3:** Develop backend logic for expense tracking, including CRUD operations for adding, editing, and deleting expenses. (Apr 14-18)
- **Task 3.4:** Implement budget goal tracking, allowing users to set and monitor financial targets. (Apr 19-23)
- **Task 3.5:** Configure Firebase Firestore for cloud-based data storage and real-time synchronization. (Apr 24-27)

Milestone 3: Fully functional backend with authentication, local storage, and cloud sync completed by the end of Week 7.

(Lutkevich, 2021)

Phase 4: Core Frontend Development (Apr 27-May 12)

- **Task 4.1:** Develop and connect login and registration screens with authentication services. (Apr 27-May 1)
- **Task 4.2:** Implement the home dashboard UI, displaying an overview of expenses, budgets, and financial trends. (May 2-4)
- **Task 4.3:** Add an expense entry form with options for receipt uploads for better financial tracking. (May 5-7)
- **Task 4.4:** Develop screens for setting budget goals and managing expense categories. (May 8-10)
- **Task 4.5:** Create an expense history screen with filtering options for easy transaction tracking. (May 11)
- **Task 4.6:** Integrate visual insights such as graphs and charts to display spending trends dynamically. (May 12)

Milestone 4: Completion of core frontend features, ensuring full functionality of key UI elements by the end of Week 10.

(Lutkevich, 2021)

Phase 5: Advanced Features (May 15-21)

- **Task 5.1:** Implement gamification features such as achievement badges and rewards to enhance user engagement. (May 15-17)
- **Task 5.2:** Add scheduled payment options and reminders to automate financial obligations. (May 18)
- **Task 5.3:** Enable multi-device synchronization using Firebase for seamless cross-platform access. (May 19-21)

Milestone 5: Completion of advanced features, making the app more interactive and user-friendly by the end of Week 12.

(Lutkevich, 2021)

Phase 6: Testing & Bug Fixing (May 22-23)

- **Task 6.1:** Conduct unit testing on individual features to verify correctness and functionality. (May 22-23)

Milestone 6: The app is fully tested, stable, and free of critical issues by the end of Week 14.

(Lutkevich, 2021)

Phase 7: Final Review & Deployment (May 27- Jun 5)

- **Task 7.1:** Conduct a final usability review to ensure all features work as intended and refine any remaining issues. (May 27-29)
- **Task 7.2:** Prepare technical documentation and a user guide to assist with onboarding and troubleshooting. (May 30- Jun 2)
- **Task 7.3:** Deploy the app for initial user testing, gathering final feedback before public release. (Jun 5)

Final Milestone: BudgetBuddy is fully developed, tested, and ready for use by the end of Week 15.

(Lutkevich, 2021)

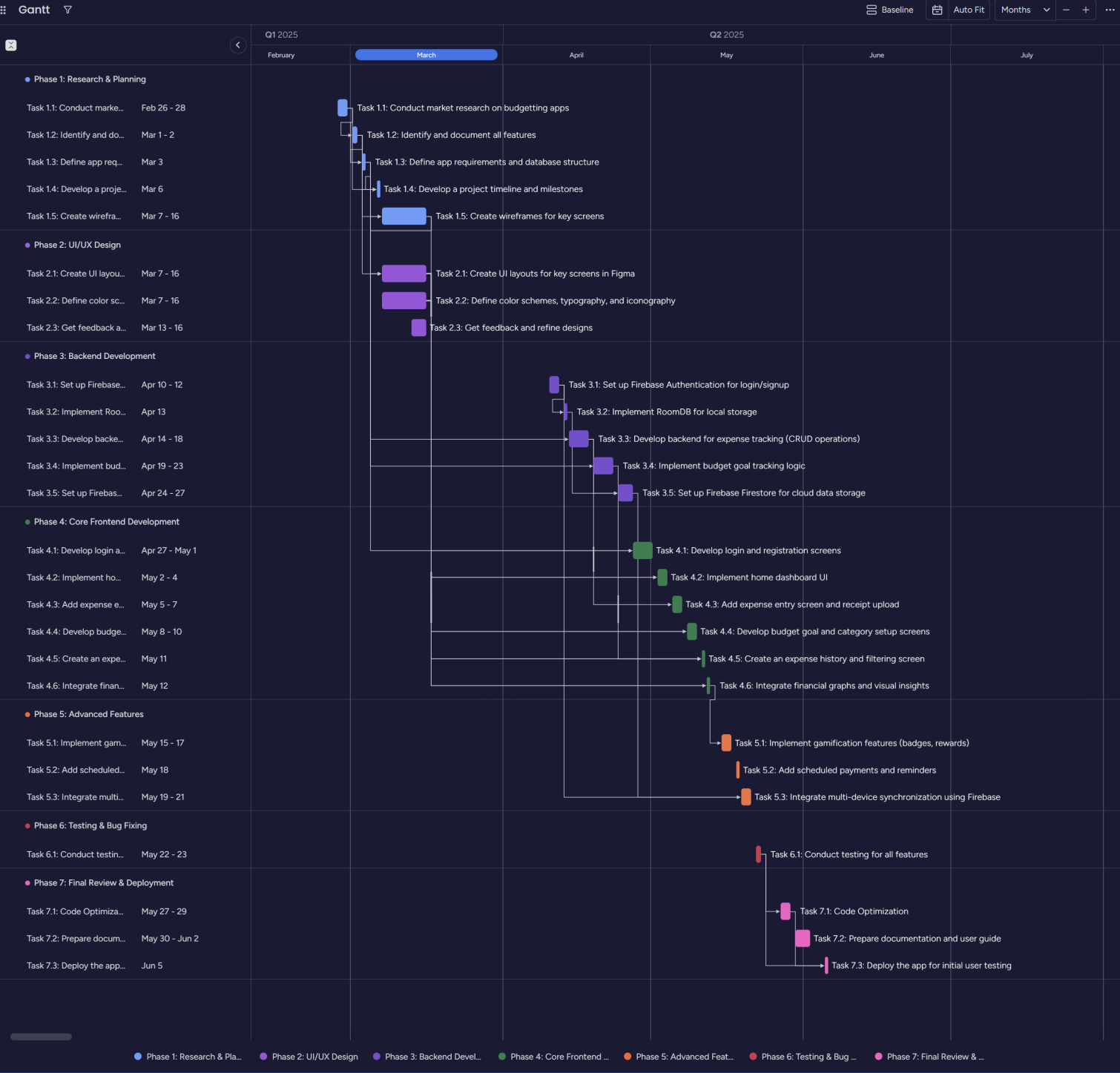
Gantt Chart

Using Monday.com to create the Gantt Chart

<https://bcadbestiess-team.monday.com/boards/1865984192>

Main Table Gantt +

New project + Add widget Search Person Filter



Conclusion

BudgetBuddy is not just a tool that helps you with budgeting, it is your personal companion that provides you thorough financial insights and encourages users to take control of financial well-being. Combining automation, real-time analytics, and interactive gamification helps the application improve user experience and encourage enhanced financial practices. Having multi-profile support, preloaded advice, and cloud syncing really ensures that everything is accessible, adaptable to various financial needs, and super flexible. The development of BudgetBuddy is guided by this design document, which combines research results with practical implementation strategies aligning with competitors within the mobile-application market. BudgetBuddy is positioned to change personal money management by clearly focusing on usability, security, and innovation, thereby making budgeting simple, intelligent, and interesting. All of the research, from the app overview and innovative features to the detailed requirements, user interface mock-ups, and project plan, will play a pivotal role in ensuring the successful development and execution of this app, ultimately delivering an efficient, user-friendly, and future-proof financial tool.

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Annexure: Disclosure of AI Usage in my Assessment

1. Section(s) within the assessment in which generative AI was used

- Section(s) within the assessment in which generative AI was used, POE PART 1 Planning and Design

2. Name of AI tool(s) used

- QuillBot Paraphrasing Tool

3. Purpose/intention behind use

- Paraphrasing, correcting grammar and spelling.

4. Date(s) in which generative AI was used

- From 27 February 2025 - 15 March 2025.

5. Link QuilBot paraphrasing tool

- <https://quillbot.com/paraphrasing-tool>